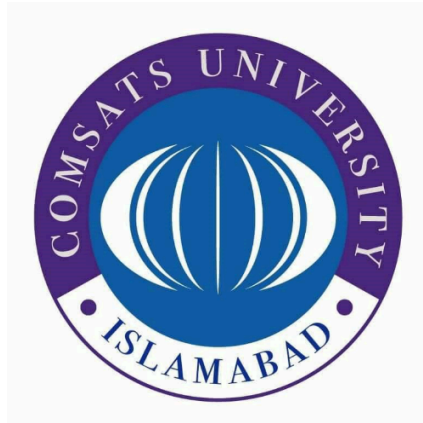




PARALLEL & DISTRIBUTED COMPUTING

PROJECT PROPOSAL

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DNA Genome Assembly using MPI and OpenMP

Brief Introduction

This project focuses on reconstructing a complete DNA genome sequence from a set of smaller DNA fragments (reads). Genome assembly is a fundamental problem in bioinformatics. We implemented two approaches for this task: a distributed web-based system using MPI (Message Passing Interface) with Flask in Python, and a high-performance parallel system using OpenMP in C++. Both versions utilize De Bruijn graph and Eulerian path algorithms to assemble the genome.

Methods / Algorithms Used

- **De Bruijn Graph:** Constructs a graph from overlapping k-mers to represent possible paths in the genome.
- **Eulerian Path:** Finds a path through the graph that visits every edge exactly once, representing a plausible sequence reconstruction.
- **Parallelization:**
 - **MPI (Python):** Uses scatter and gather for parallel graph building across ranks.
 - **OpenMP (C++):** Uses parallel for loops and thread-safe graph construction.

Programming Languages / Tools

- Python with Flask and mpi4py for the web-based parallel app
- C++ with OpenMP for fast multithreaded genome assembly
- HTML/CSS and Tailwind CSS for the frontend UI
- MPI Libraries: MPICH / OpenMPI
- g++ with OpenMP support

Conclusion

Through the combination of MPI and OpenMP, we demonstrated efficient and scalable genome assembly. The web app offers accessibility and distribution, while the C++ version delivers speed. Our project highlights the importance of parallel computing in solving computational biology problems efficiently.

